Supplementary Information:

Context Without Contact: Top-Down Information Flow from Shared Modulation in Uncoupled Prebiotic Systems

Celia Blanco[1,2]

1. Centro de Astrobiología (CSIC-INTA), Ctra de Torrejón a Ajalvir, km 4, 28850 Torrejón de Ardoz, Madrid, Spain

2. Blue Marble Space Institute of Science, Seattle WA, 98104 USA

ORCID: 0000-0003-1536-1493

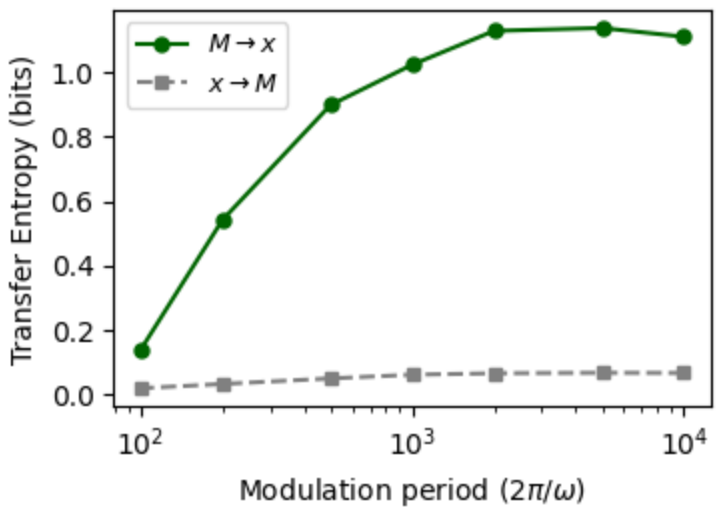
Corresponding author: Celia Blanco, celia.blanco@bmsis.com

This PDF file includes:

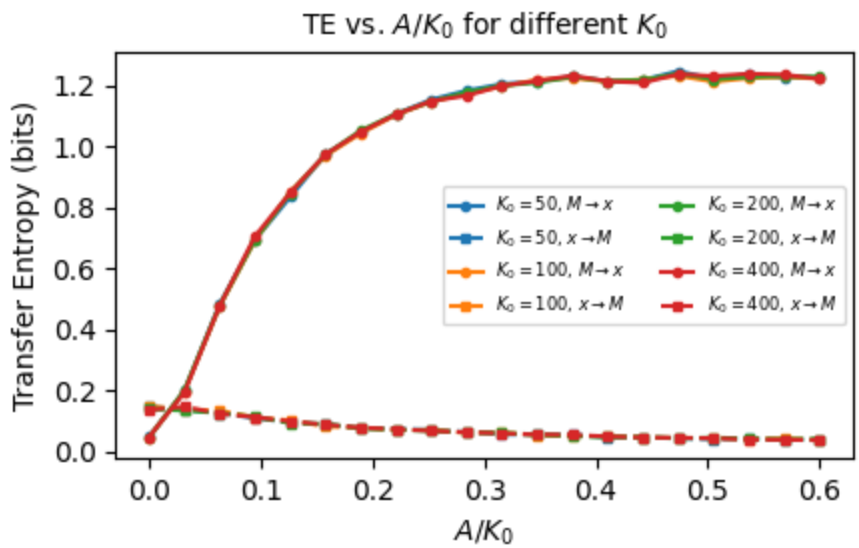
Supporting Figures S1 to S5

Other supporting materials for this manuscript include the following:

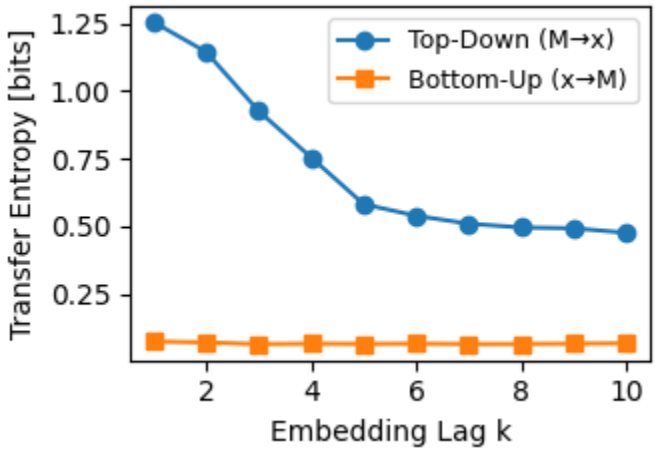
Code used in the modeling and subsequent analysis is available at https://github.com/celiablanco/macro_to_micro



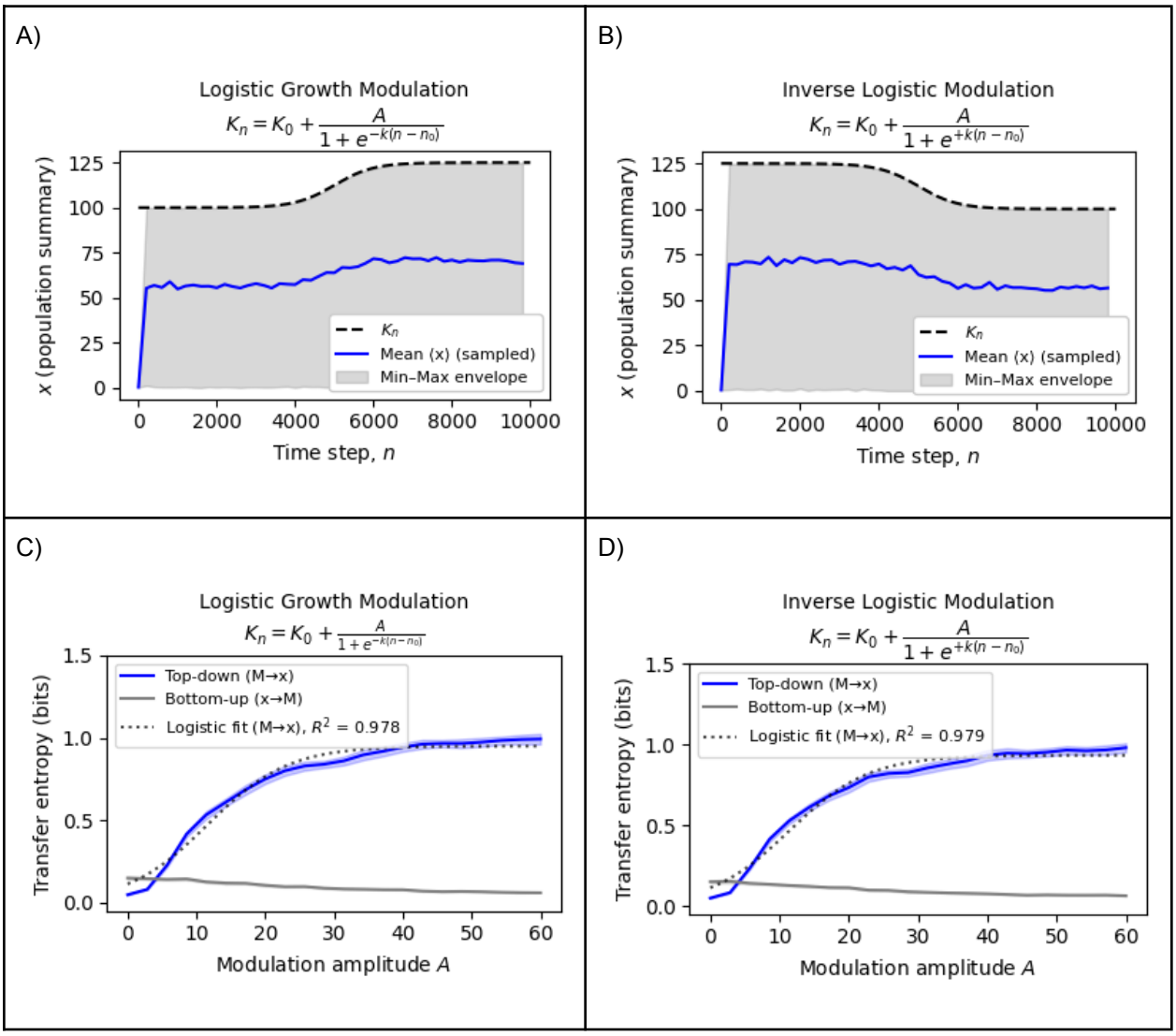
Supplementary Figure S1. Dependence of macro-to-micro transfer entropy on modulation frequency. Transfer entropy computed in both directions, $TE(M \to x)$ and $TE(x \to M)$, at fixed modulation amplitude $A = 25$ as a function of the modulation period $2\pi/\omega$. Results are averaged over 10 independent simulations ($N = 1,000$, $K_0 = 100$, $r_i = [3.9, 4.0]$, $T = 10,000$ steps, first 1,000 discarded). $TE(M \to x)$ remains high for slow modulation (long periods) but decreases for faster oscillations, indicating that rapid environmental fluctuations average out relative to the intrinsic chaotic dynamics and therefore impose weaker coherent structure across the population. In contrast, $TE(x \to M)$ remains small across modulation periods.



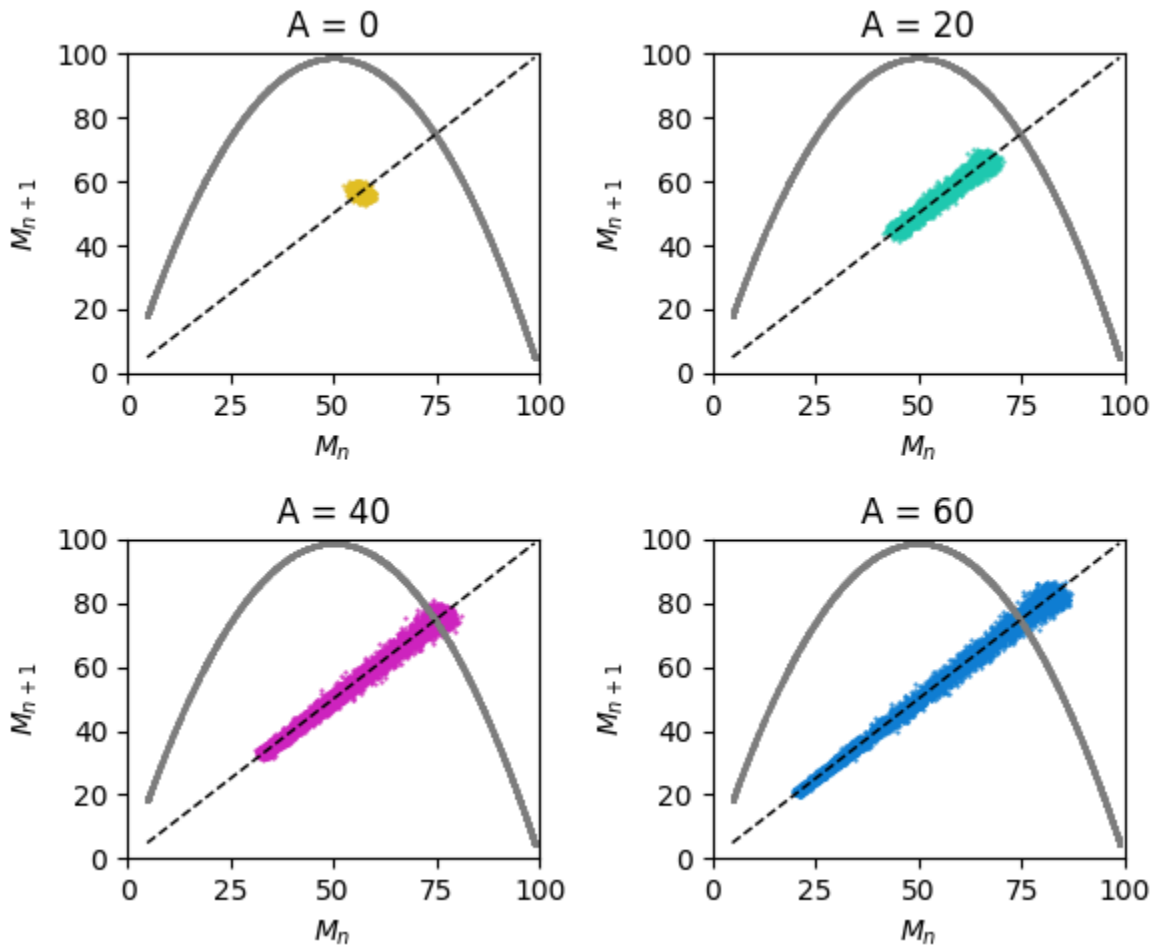
Supplementary Figure S2. Transfer entropy as a function of relative modulation amplitude A/K_0 for different baseline carrying capacities. Transfer entropy in both directions, $TE(M \to x)$ and $TE(x \to M)$, shown as a function of the ratio A/K_0 for K_0 = 50, 100, 200, and 400. The approximate collapse of the curves indicates that the relative modulation amplitude acts as the relevant control parameter for the growth of macro-to-micro transfer entropy. $TE(x \to M)$ remains small across all values of A/K_0. Results are averaged over 5 seeds (*N* = 1,000, $K_0 = 100$, $r_i \in [3.9, 4.0]$, and *T* = 10,000 steps, with first 1,000 discarded, $\omega = 2\pi/2000$, embedding lag $k = 2$, 15 bins).



Supplementary Figure S3. Mean transfer entropy (TE) as a function of embedding lag k for modulation amplitude A=25. Top-down TE (M→x, circles) measures information from the macroscopic mean field to an individual unit, while bottom-up TE (x→M, squares) measures the reverse direction. Values are averaged over 10 random seeds. Parameters: N=1000, K_0=100, $r_i \in [3.9,4.0]$, T=10,000 steps (first 1,000 discarded), 15-bin histograms.



Supplementary Figure S4. Directional information flow from macroscopic to microscopic dynamics under different modulation profiles. Transfer entropy (TE) between the population mean M_n and a representative unit x_n, computed in both directions: $M \to x$ (blue) and $x \to M$ (gray), for a population of N = 1000 uncoupled logistic units evolving under carrying capacity modulated by (A,C) logistic increase, and (B, D) inverse logistic. In all cases $K_0 = 100, A = 25, r_i \in [3.9, 4.0]$, and steepness is $k = 0.05$, simulated over 10 independent simulations, each run for T = 10,000 steps (first 1,000 discarded). Top row: The black curve shows the external modulation K_n, the blue line corresponds to the population mean <*x*>, sampled every 200 steps, and the shaded gray band indicates the range of individual states at each sampled point. Bottom row: TE estimated using a histogram-based method with lag k = 2 and 15 bins. Shaded regions indicate 95% confidence intervals.



Supplementary Figure S5. Structural modulation induces low-dimensional macroscopic dynamics. Return maps of (M_n, M_{n+1}) for $A = \{0, 20, 40, 60\}$, N = 1,000, $K_0 = 100$, $r_i \in [3.9, 4.0]$, and T = 10,000 steps (first 1,000 discarded). The dashed line shows the identity $M_{n+1} = M_n$ and the gray curve shows the single-unit logistic map for comparison. At A=0, points cluster near ($K_0/2$, $K_0/2$). As A increases, shared modulation widens the distribution of M_n, producing a diagonal spread.